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School of Engineering

Department of Industrial and Systems Engineering

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Course Project

Team: Future Forecasters

Report #1

Contribution	
Member	Contribution
AbdelRahman	EDA, Report Writing (Intro, Data Analysis)
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1. Introduction & Problem Statement

Forecasting electricity prices has become increasingly of interest due to the liberalization of energy markets, the integration of renewable energy sources, and the instability of fuel prices over the years. Hence, an accurate short-term electricity price predictions are essential for the market stakeholders to optimize bidding strategies, ensure grid reliability, and manage operational risks (Weron, 2014). In this forecasting study, the Locational Marginal Pricing (LMP), which reflects the marginal cost of delivering electricity at specific locations, is the primary target variable of interest. This study aims to develop a 24-hour ahead forecasts for LMP using time series modeling techniques and exogenous predictions, which are load forecasts, solar forecasts, wind forecasts, fuel price, and air temperature.

To meet this objective, the study utilizes a multi-step modeling pipeline. The forecasting framework begins with the modeling of fuel price variable using a Naïve time series model that is trained on over three years of historical data, “Train_Round2”. By investigating the correlation matrix (heat map) of target and predictor variables, it can be seen that fuel price is itself a key predictor of LMP; however, it is partially unavailable in the testing dataset. Hence, a separate forecasting step ensures its availability before proceeding to LMP modeling. The LMP is forecasted using a Seasonal ARIMA with Exogenous Regressors (SARIMA-X) model. This enables the model to incorporate time-lagged features with multiple predictor variables. This modeling framework will be effective in capturing both seasonality and the influence of exogenous predictors on electricity prices. Additionally, two benchmark models, the Naïve and Seasonal Naïve with a 24-hour lag, are used to establish baseline performance and validate the current model. Forecast accuracy is evaluated using Mean Absolute Error (MAE).

2. Exploratory Data Analysis (EDA)

2.1. Time Series Visualization and STL Decomposition Plots (Daily)

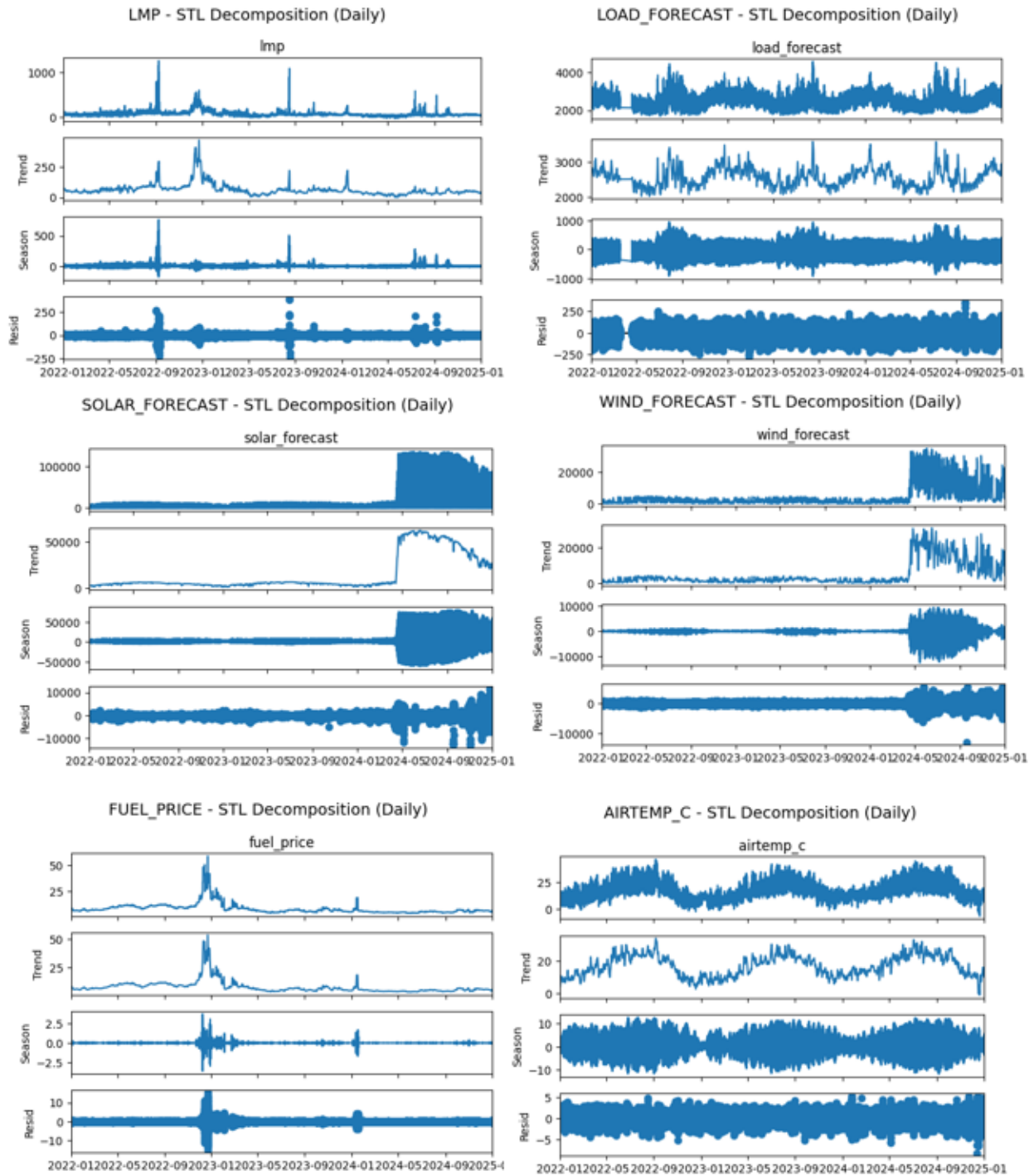


Figure 1 STL (Seasonal-Trend decomposition using Loess) daily decomposition plots of all variables: (a) Top Left: STL for LMP, (b) Top Right: STL for Load Forecast, (c) Middle Left: STL for Solar Forecast, (d) Middle Right: STL for Wind Forecast, (e) Bottom Left: STL for Fuel Price, (f) Bottom Right: STL for Air Temperature

2.2. ACF Plots

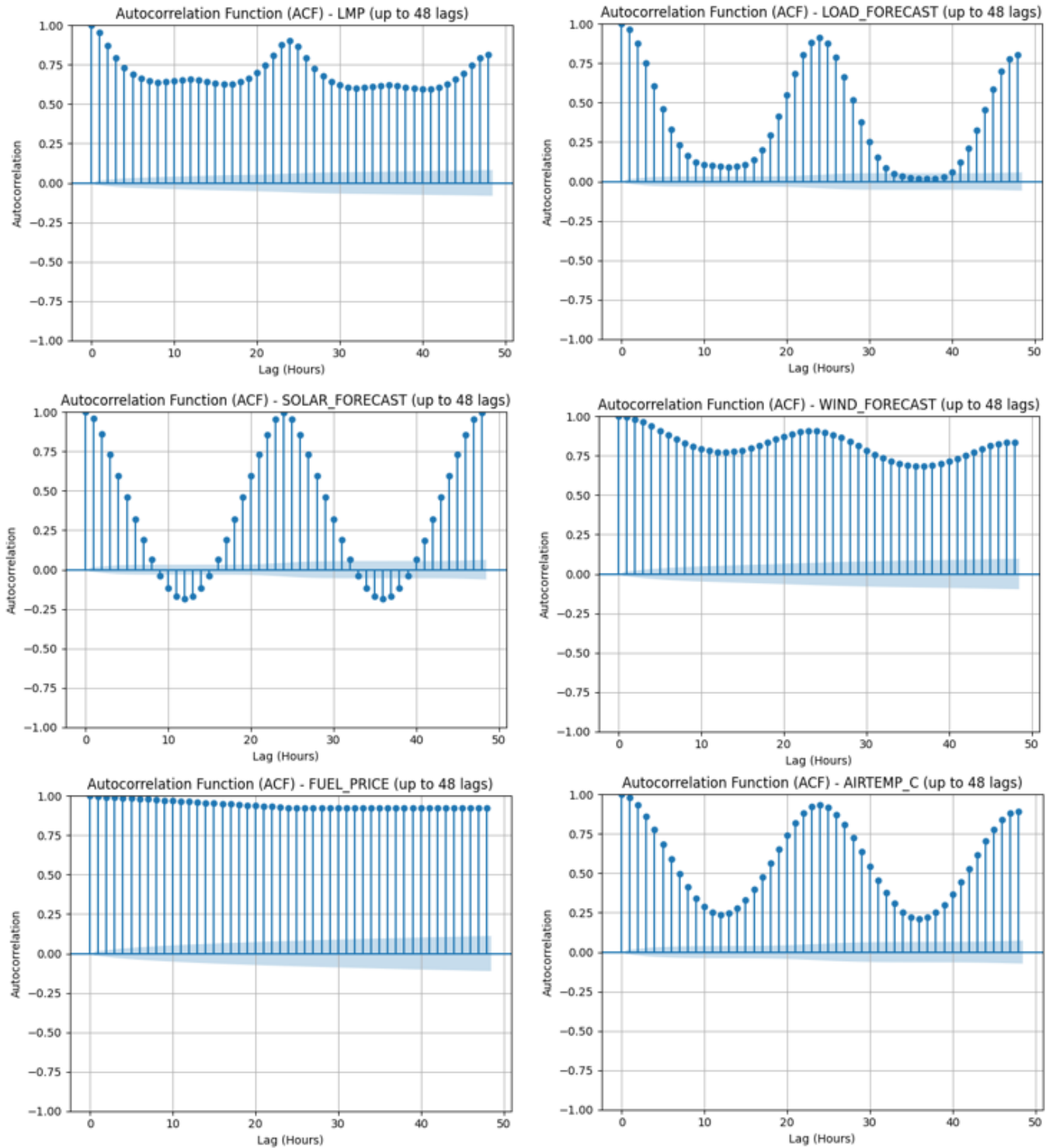


Figure 2 ACF Plots (up to 48 lags – 2 days) for all variables: (a) Top Left: ACF for LMP, (b) Top Right: ACF for Load Forecast, (c) Middle Left: ACF for Solar Forecast, (d) Middle Right: ACF for Wind Forecast, (e) Bottom Left: ACF for Fuel Price, (f) Bottom Right: ACF for Air Temperature

The STL decomposition plots, in figure 1, displays the raw signal, trend, seasonal, and residual components on a daily periodicity for all variables in the dataset. The plots reveal key insights into the behavior of each variable, which inform our modeling choices. For LMP, strong irregularities and spikes in the residual component suggest instability and the presence of outliers, as shown in figure 1 (a), which the model must accommodate. The load forecast decomposition displays a clear seasonal pattern and consistent daily trend behavior, as shown in figure 1 (b). These behaviors indicate that this variable could significantly contribute to improving LMP forecasts when included as an exogenous variable. In contrast, the solar forecast and wind forecasts decompositions exhibit abrupt structural changes around early 2024, as shown in figures 1 (c) and (d). The fuel price shows a relatively smooth time series with an irregular spike at early 2023 and a low-variance dynamics, shown in figure 1 (e). Lastly, air temperature displays a well-defined seasonal pattern with moderate variability, making it a stable exogenous regressor, shown in figure 1 (f). From the STL plots in figure 1, the fuel price raw time series appears closest to being stationary, with a relatively flat trend and minor seasonal variation and an abnormal spike in early 2023 and 2024. This makes this predictor suitable for a Naïve model. In contrast, LMP, load forecast, and air temperature show non-stationary behaviors due to strong seasonal trends and varying means over time. For these time series, seasonal differencing or transformation are required to induce stationarity prior to model fitting.

The ACF plots for all variables included in the model, shown in figure 2, show insights for the model formulation by analyzing the autocorrelation structures present in each time series for up to 48 lags (or 2 days). Most notably, the LMP, figure 2 (a), and load, figure 2 (b), forecast exhibit a strong 24-hour cyclical autocorrelations. This indicates a definite daily seasonality, thus justifying the use of seasonal terms in the forecasting model. The solar forecast, figure 2 (c), and air temperature, figure 2 (f), ACFs show a sinusoidal behavior with clear daily cycles. This suggests their potential utility as seasonally structured exogenous regressors. In contrast, fuel price, figure 2 (e), ACF plot displays a persistently high autocorrelation across all lags, implying strong temporal dependence but minimal short-term fluctuation. The wind forecast, figure 2 (d), ACF plot shows moderate but gradually decaying autocorrelations, suggesting short-term memory but without strong daily periodicity.

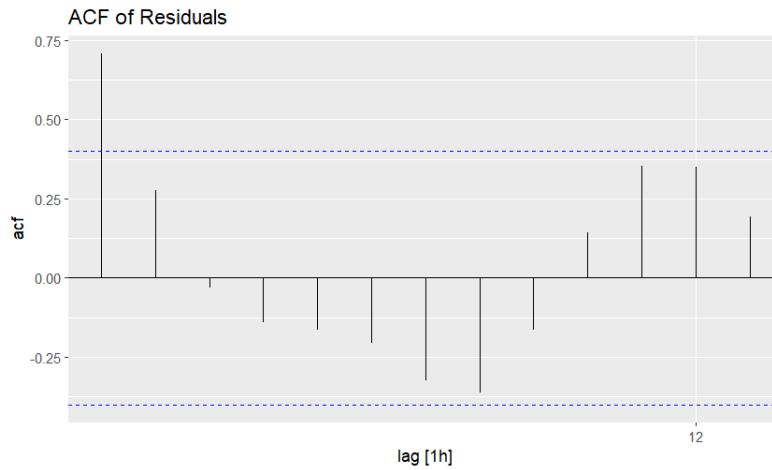
3. Models & Methodology

Data analysis motivated the implementation of a linear time series regression with SARIMA errors (SARIMA-X) model to forecast the LMP. Specifically, the model that is fit based on the extensive training set was a linear model regressed on forecasts of Load, Solar, Wind, Fuel Price, and Air Temperature. The errors are forecasted using $ARIMA(3,0,1)(0,1,1)_{24}$. The linear model captures the relationship between the exogenous variables and the LMP while the SARIMA portion captures the strong seasonality present in the time series and accounts for the non-stationary behavior of the LMP time series. Note that the seasonal period is set to 24 hours as the data has strong daily seasonality. Also note the first degree seasonal differencing used to make

the time series more stationary. Another key aspect to the model is that it uses Fuel Price to forecast the LMP. Since a fuel price forecast is not already provided, the naïve model is used to forecast the fuel price. The naïve model suites fuel price well due to its relatively low volatility and constant behavior over each day.

4. Validation & Experiments

To validate the model, both ex-post and ex-ante forecasting is performed. Ex-post forecasting is completed using the actuals from 1/2/2025. The SARIMA-X model results in an MAE of 3.91 which is well above the benchmark of 5.99 of the naïve model and even above the seasonal naïve (4.43). In the same way, experiments are performed testing different models for fuel price, but the naïve model is shown to perform the best. Additionally, residual analysis is also conducted using similar techniques as initial data analysis (plotting, histogram and ACF). The plots reveal that while the model captures some of the trend and seasonality there is still room for improvement. While all but one residual ACF falls below the significance threshold, there is still somewhat of a cyclic pattern represented, indicating that the model has not captured all of the



seasonality. Furthermore, the histogram of residuals (not pictured) reveals that they are not centered around zero suggesting further improvement is possible.

Likewise, ex-ante forecasting is performed using the reserved training data and an MAE of 2.846 is calculated (comparable to ex-post).

Figure 3 ACF Plots (up to 12 lags) for residuals of the ARIMA-X model during ex-post forecasting of 1/2/25

5. Conclusion

This study presents a SARIMA-X modeling framework for day-ahead LMP forecasting using a naïve fuel forecast model. The results demonstrate that the proposed SARIMA-X model significantly outperforms benchmarks of the naïve and seasonal naïve model in ex-post forecasting. Residual diagnostics show that there is room for improvement, as some cyclic behavior remains. With that said, the modeling framework provides a promising foundation for electricity price forecasting that can be further enhanced with additional predictive features.

References

- [1] Weron, R. (2014). *Electricity price forecasting: A review of the state-of-the-art with a look into the future*. International Journal of Forecasting, 30(4), 1030–1081.
<https://doi.org/10.1016/j.ijforecast.2014.08.008>